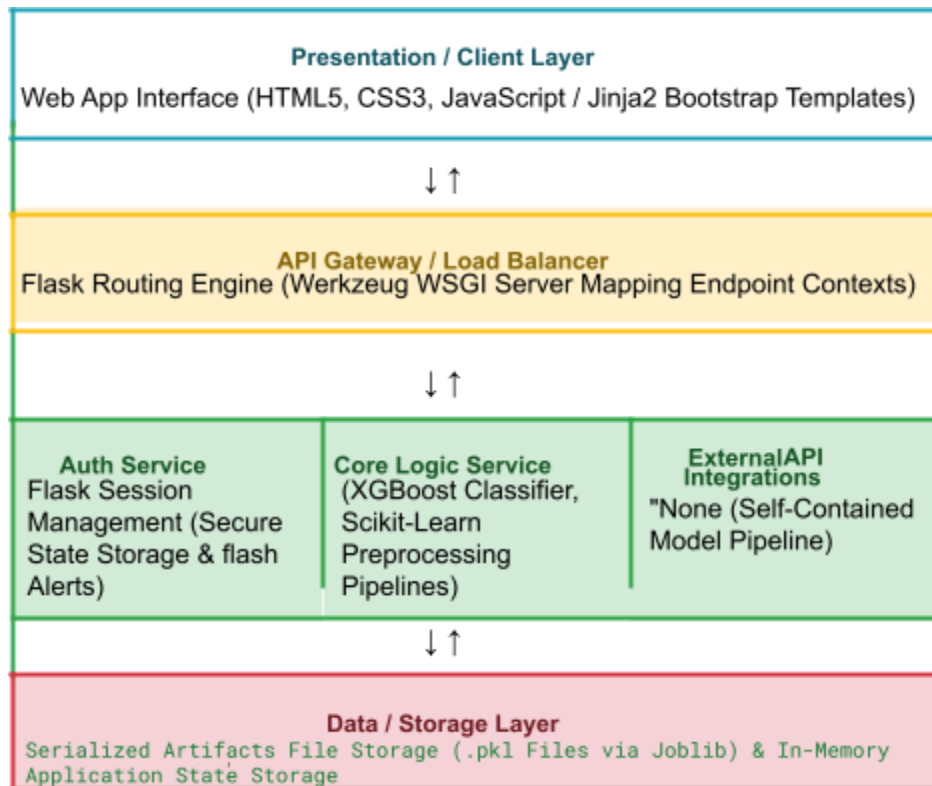


Solution Architecture

Date	4 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	[Renders the user interface where loan/credit card applicants or bank staff input customer credit details (income, credit score, history) and view the final automated approval decision.]	[HTML5, CSS3, JavaScript / Jinja2 Templates]
[API Gateway]	[Manages incoming HTTP requests from the front-end application, routes them to the correct application endpoints, and returns JSON-formatted prediction results back to the client.]	[Flask Routing Engine, Werkzeug WSGI Server]
[Microservices]	[Handles input data sanitization, categorical feature encoding, and scales numerical records. It loads the serialized machine learning model to execute real-time binary risk scoring (Approved / Rejected).]	[Python, Flask, scikit-learn, pandas, numpy]
[Database]	[Stores structured historical customer data and application history records utilized for model training, verification checks, and performance logging].	[Local File System (CSV format)]